

EMOS Tutorial Exercises

Exercise 1 — Stability Consensus Analysis with a Lambda Function

Context

A database such as the Materials Project reports thermodynamic stability through the convex hull distance, computed from DFT calculations using a specific exchange-correlation functional and a carefully curated set of elemental reference energies. A machine-learned interatomic potential (MLIP) such as MatterSim was trained on a different dataset and uses different energy references. When you ask both sources whether a material is stable, they may disagree — not because one is simply wrong, but because stability is defined relative to a reference state, and those reference states differ. A material that looks stable under one framework but not another is a poor basis for expensive downstream experiments. The challenge is to identify which materials earn a consistent verdict across independent computational sources, and to flag those that do not. In this exercise you examine five well-known semiconductors — Si, GaAs, GaN, ZnO, and TiO₂ — chosen because their experimental stability is not in doubt, so any disagreement you find is an artefact of the methods rather than the materials.

Task and Deliverables

Build the pipeline below in the node editor. Place a **Materials Project** database IU and configure it to query the five compounds by chemical formula. Wire its output CIFs and JSON into a **MatterSim** predictor IU set to return `formation_energy`. Wire the MatterSim output into a **Lambda Function** node and paste the code provided below into the text box. The lambda reads each material's `hull_distance` (reported by Materials Project in the JSON) and MatterSim's predicted `formation_energy`, then annotates each entry with three new fields: `db_stable`, `mlip_stable`, and `consensus`. Wire the lambda output into a **Text Viewer** and a **CIF Viewer**.

Deliverables:

1. A screenshot of the completed node editor pipeline, with all node labels and wire connections visible.
2. A table copied from the Text Viewer listing each material's `hull_distance`, `formation_energy`, `db_stable`, `mlip_stable`, and `consensus` values.

3. A written commentary (150–200 words) identifying which materials, if any, produced consensus = False, and proposing a reason. Consider: do the two methods use the same elemental reference energies? Does a negative formation energy guarantee a material sits on the convex hull? Note explicitly any limitations of using $\text{formation_energy} < 0$ as a proxy for hull stability.

Exercise 2 — The FET Material Selection Pyramid

Context

A field-effect transistor (FET) channel material must satisfy several simultaneous requirements, and almost no material satisfies all of them. The band gap must lie in the range $0.9 \leq E_g \leq 4.0$ eV: too small and the off-state leakage current becomes unacceptable; too large and the drive current in the on-state drops. Good electrostatic gate control requires a reasonably high dielectric constant, $\kappa \geq 3.0$. Fast switching requires adequate carrier transport: electron mobility $\mu_e \geq 50$ cm²/(V·s) is a practical lower bound. Finally, only materials near the convex hull — $E_{\text{hull}} \leq 0.25$ eV/atom — are realistically synthesisable. The challenge is that inorganic crystal databases contain hundreds of thousands of entries and most fail at least one of these criteria. Applying all filters simultaneously to the raw database is wasteful because the most expensive computations (structural relaxation) would be run on materials that a cheap band gap check would have eliminated immediately. The pyramid strategy applies the cheapest filter first — a database query — and reserves costly predictors for the small survivor pool, minimising total computation time.

Task and Deliverables

Build a three-stage pipeline. **Stage 1** places a **JARVIS-DFT** database IU and an **Alexandria** database IU side by side, both queried for binary and ternary compositions. Wire both outputs into a **Filter Function** node with the condition $\text{band_gap} \geq 0.9 \ \& \ \text{band_gap} \leq 4.0 \ \& \ \text{hull_distance} \leq 0.25$. Record the surviving count before proceeding. **Stage 2** wires those CIFs into a **GBFS** predictor IU requesting band_gap , $\text{dielectric_constant}$, and electron_mobility , then into a second **Filter Function** with $\text{band_gap} \geq 0.9 \ \& \ \text{band_gap} \leq 4.0 \ \& \ \text{dielectric_constant} \geq 3.0 \ \& \ \text{electron_mobility} \geq 50$. Record the new survivor count. **Stage 3** wires the output into a **MatterSim** predictor IU configured to perform structural relaxation, then into a third **Filter Function** with $\text{max_residual_force} \leq 0.10$. End with a **Text Viewer** and a **CIF Viewer**.

Deliverables:

1. A screenshot of the full pipeline with all three filter nodes visible and their condition strings legible.
2. A table or bar chart showing the candidate count at each pyramid stage (initial retrieval, after Stage 1, after Stage 2, after Stage 3).
3. A written commentary (150–200 words) identifying at which stage the largest reduction occurs and why, and assessing whether any surviving material is surprising. If a material appears in the final set that you did not expect, look it up in the Materials Project and check whether the GBFS-predicted mobility is consistent with the literature value. Note that GBFS is a surrogate model and its predictions carry uncertainty not reflected in these point estimates.

Exercise 3 — Extending the Pyramid with Synthesizability Screening

Context

Thermodynamic stability on the convex hull tells you that a material will not spontaneously decompose into competing phases at absolute zero under idealised conditions. It says nothing about whether a chemist can actually make it. Real synthesis involves reaction kinetics, precursor availability, processing temperature and atmosphere, and substrate compatibility — none of which are captured by a formation energy. SynthNN is a neural network trained on the compositions of experimentally reported versus purely theoretical compounds; it returns a synthesizability score between 0 and 1, where higher values indicate closer resemblance to compositions that have been successfully made in a laboratory [Antoniuk et al., *npj Computational Materials*, 9, 155, 2023]. The EMOS MOSFET case study used a threshold of 0.70 to isolate the two most actionable candidates from a pool of eight structurally plausible materials, arriving at PbI_2 and VO_2 . The important caveat is that SynthNN scores are composition-based: two polymorphs of the same formula receive the same score, so the filter cannot distinguish between a well-known stable polymorph and a hypothetical high-pressure phase of identical composition.

Task and Deliverables

Take the pipeline from Exercise 2 and extend it. After the Stage 3 MatterSim filter, add a **SynthNN** predictor IU and wire its output into a fourth **Filter Function** node with the condition `synthesizability_score >= 0.70`. This constitutes Stage 4. Wire the Stage 4 output into a **Text Viewer** and a **CIF Viewer**. Run the extended pipeline and compare the final output with the Stage 3 survivor list from Exercise 2.

Deliverables:

1. A screenshot of the extended pipeline clearly showing the SynthNN node and Stage 4 filter appended to the Exercise 2 graph.
2. An updated table showing candidate counts at all four stages.
3. A written commentary (200–250 words) addressing: which materials, if any, passed structural relaxation but were removed by the synthesizability filter? What does this imply about using Ehull alone as a proxy for experimental accessibility? Consider also the polymorph limitation noted above — if a material is removed by the synthesizability filter, can you tell from the score alone whether this is because the composition is genuinely inaccessible or because the specific polymorph is unusual even if the composition is common?

Exercise 4 — Screening for LED Materials Using a Lambda-Computed Emission Wavelength

Context

Light-emitting diodes (LEDs) and laser diodes convert electrical energy directly into photons. The emitted photon energy is set, to a first approximation, by the semiconductor band gap: a carrier recombination event releases energy E_g , producing a photon of wavelength $\lambda = hc/E_g$. In practical units this becomes $\lambda \text{ (nm)} = 1240/E_g \text{ (eV)}$. For visible-light applications, the relevant band gap range maps to the visible spectrum: blue light (450–495 nm) requires E_g between approximately 2.5 and 2.8 eV, green (495–570 nm) requires 2.2–2.5 eV, and red (620–750 nm) requires 1.65–2.0 eV. This simple relation is a starting point, not a complete design rule — it ignores whether the gap is direct or indirect, defect tolerances, and the host matrix's optical absorption — but it is sufficient to narrow a database of hundreds of thousands of compounds to a manageable shortlist. The challenge is that the EMOS property dictionary does not contain a pre-computed `emission_wavelength_nm` field: you must derive it from `band_gap` inside the pipeline. This is exactly the use case for a Lambda Function node.

Task and Deliverables

Build the following pipeline. Place a **JARVIS-DFT** database IU configured to retrieve compounds with `band_gap` between 1.6 and 3.1 eV and `formation_energy` below 0 eV/atom. Wire its output into a **Filter Function** with `band_gap >= 1.6 & band_gap <= 3.1 & formation_energy < 0.0 & hull_distance <= 0.2` to remove unstable entries early. Wire the output into a **Lambda Function** node with the code below, which computes `emission_wavelength_nm` and appends it to each material's JSON record. Wire the lambda output into a second **Filter Function** where you select your target colour — choose one from the three ranges listed in the context paragraph and enter the

corresponding condition, for example `emission_wavelength_nm >= 450 & emission_wavelength_nm <= 495` for blue. End with a **Text Viewer** and a **CIF Viewer**.

Deliverables:

1. A screenshot of the pipeline with the Lambda Function node selected, showing the code text box and the colour-target Filter Function condition string.
2. A table from the Text Viewer listing each surviving material's formula, `band_gap`, and computed `emission_wavelength_nm`.
3. A written commentary (200–250 words) evaluating the candidate list. Look up whether any surviving compounds are known LED or phosphor materials in the literature. Identify at least one limitation of the $\lambda=1240/E_g$ approximation — specifically, consider what happens if the band gap is indirect, and why this matters for radiative recombination efficiency. Note also that JARIS-DFT band gaps are computed using semi-local DFT, which systematically underestimates experimental optical gaps, and discuss the direction in which this shifts your predicted wavelengths.